

Quadcopter Flight Dynamics:

Vector Basis Transformation vs. Rigid Body Physics

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Abstract

This document rigorously analyzes two control strategies for an X-configuration quadcopter. We first mathematically prove “Idea 1”: that controlling diagonal motor pairs using two PIDs on a 45° rotated basis frame works purely geometrically. We then introduce Newtonian and Eulerian physics to prove why this method catastrophically fails in reality due to reactive torque and gyroscopic precession. Finally, we derive the “Idea 2”: using three decoupled PIDs (Roll, Pitch, Yaw) combined with a Control Allocation Matrix (Mixer) to mathematically isolate physical forces and achieve stable flight.

1 System Definition & The “4-Ball” Model

Let us define the quadcopter as a rigid body with 4 rotors (the “4 balls”) situated at a distance l from the center of mass (COM). We use a standard right-handed coordinate system where X is Forward, Y is Left, and Z is Up.

- **Motor 4 (Front-Left, FL):** Clockwise (CW) rotation.
- **Motor 2 (Front-Right, FR):** Counter-Clockwise (CCW) rotation.
- **Motor 3 (Back-Left, BL):** Counter-Clockwise (CCW) rotation.
- **Motor 1 (Back-Right, BR):** Clockwise (CW) rotation.

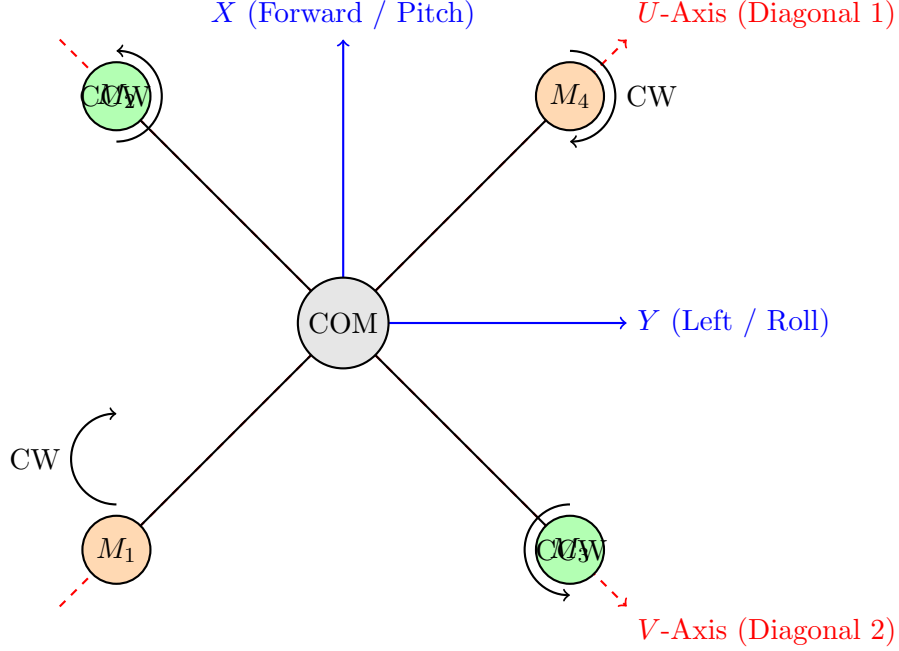


Figure 1: The Quadcopter model showing Standard Axes (X, Y) and Idea 1 Tilted Axes (U, V)

2 Idea 1: The Theoretical Math (Diagonal Splitting)

In Idea 1, we assume we can control the drone by tilting its axes by 45° , converting the X (Pitch) and Y (Roll) commands into U and V diagonal commands.

2.1 Coordinate Frame Rotation (Basis Transformation)

To map a desired pitch (forward) and roll (left) vector into our new diagonal motor pairs, we apply a 45° rotation matrix $R(\theta)$:

$$\begin{bmatrix} U \\ V \end{bmatrix} = \begin{bmatrix} \cos(45^\circ) & \sin(45^\circ) \\ -\sin(45^\circ) & \cos(45^\circ) \end{bmatrix} \begin{bmatrix} X_{target} \\ Y_{target} \end{bmatrix} \quad (1)$$

Because $\cos(45^\circ) = \sin(45^\circ) = \frac{\sqrt{2}}{2}$, we get:

$$U_{target} = \frac{\sqrt{2}}{2}(X_{target} + Y_{target}), \quad V_{target} = \frac{\sqrt{2}}{2}(X_{target} - Y_{target}) \quad (2)$$

2.2 The 2-PID Control Law

We assign two PIDs to these mathematical axes:

- **PID_u**: Corrects the U -axis by controlling the M_4 & M_1 diagonal pair.
- **PID_v**: Corrects the V -axis by controlling the M_2 & M_3 diagonal pair.

Theoretical Conclusion: Purely in linear algebra (assuming superposition of thrust vectors), this works perfectly. If we want to pitch forward ($X_{target} = 1, Y_{target} = 0$), then $U = 0.707$ and $V = 0.707$. Both diagonals increase equally, lifting the back and lowering the front. The math is theoretically flawless.

3 The Physics Reality: Why Idea 1 Fails

When we introduce real-world physics (Newton-Euler Rigid Body Dynamics), Idea 1 catastrophically fails. We rank the physical phenomena responsible.

3.1 Rank 1: Reactive Torque Imbalance (The Yaw Spin) - 80% of Failure

According to Newton’s Third Law, every motor generates aerodynamic drag (torque) in the opposite direction of its spin. Let k_m be the torque constant. The total Yaw torque (τ_ψ) on the drone is:

$$\tau_\psi = k_m(\Omega_4^2 + \Omega_1^2) - k_m(\Omega_2^2 + \Omega_3^2) \quad (3)$$

Where Ω is motor angular velocity. Notice that Ω_4 and Ω_1 are the ***U-Axis motors (Both CW)***.

The Crash Scenario: A small gust of wind tilts the drone on the *U*-Axis. PID_u increases Ω_4 and Ω_1 to level the drone.

$$\tau_\psi = k_m(\text{Large Number}) - k_m(\text{Small Number}) \neq 0$$

Result: The drone generates massive uncompensated Yaw torque. It instantly spins like a top. The physical position of the motors no longer aligns with the PID’s internal math, resulting in instant feedback failure.

3.2 Rank 2: Gyroscopic Precession (The Wobble) - 15% of Failure

A spinning propeller is a gyroscope. When a gyroscope is tilted, it exerts a reactive torque 90° out of phase. Let J_r be rotor inertia, and ω_{body} be the drone’s rotational rate vector $[\dot{\phi}, \dot{\theta}, \dot{\psi}]^T$. The gyroscopic torque is:

$$\vec{\tau}_{gyro} = \sum_{i=1}^4 J_r \left(\vec{\omega}_{body} \times \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} (-1)^{i+1} \Omega_i \right) \quad (4)$$

The Crash Scenario: If PID_u tilts the drone purely on the *U*-Axis, precession dictates a reactive force is pushed 90° away—directly onto the *V*-Axis. PID_v detects a disturbance and fires its motors, inducing precession back onto the *U*-axis. **Result:** A chaotic, divergent feedback loop known as “Cross-Coupling” or the “Toilet Bowl Effect.”

4 Idea 2: The Mathematical Decoupling

To fix these physics problems, we abandon controlling physical diagonals. Instead, we create **3 distinct PIDs** governing abstract physical rotations: Roll (ϕ), Pitch (θ), and Yaw (ψ).

4.1 Deriving the Newton-Euler Matrix

We express the physical forces on the drone as a function of the squares of motor speeds (Ω_i^2). Let k_f be the thrust constant and l the arm length.

$$\text{Total Thrust } (T) = k_f(\Omega_1^2 + \Omega_2^2 + \Omega_3^2 + \Omega_4^2) \quad (5)$$

$$\text{Roll Torque } (\tau_\phi) = l \cdot k_f(\Omega_3^2 + \Omega_4^2 - \Omega_1^2 - \Omega_2^2) \quad (6)$$

$$\text{Pitch Torque } (\tau_\theta) = l \cdot k_f(\Omega_1^2 + \Omega_3^2 - \Omega_2^2 - \Omega_4^2) \quad (7)$$

$$\text{Yaw Torque } (\tau_\psi) = k_m(\Omega_1^2 + \Omega_4^2 - \Omega_2^2 - \Omega_3^2) \quad (8)$$

We write this in matrix form, $\vec{U} = A\vec{\Omega}^2$:

$$\begin{bmatrix} T \\ \tau_\phi \\ \tau_\theta \\ \tau_\psi \end{bmatrix} = \begin{bmatrix} k_f & k_f & k_f & k_f \\ -lk_f & -lk_f & lk_f & lk_f \\ lk_f & -lk_f & lk_f & -lk_f \\ k_m & -k_m & -k_m & k_m \end{bmatrix} \begin{bmatrix} \Omega_1^2 \\ \Omega_2^2 \\ \Omega_3^2 \\ \Omega_4^2 \end{bmatrix} \quad (9)$$

4.2 The Control Allocation Matrix (The “Mixer”)

We invert the physical matrix A to find out exactly how much power to give each motor based on the abstract PID outputs. $\vec{\Omega}^2 = A^{-1}\vec{U}$. Simplifying the constants for the flight controller:

$$\begin{bmatrix} \Omega_1^2 \\ \Omega_2^2 \\ \Omega_3^2 \\ \Omega_4^2 \end{bmatrix} = \begin{bmatrix} 1 & -1 & 1 & 1 \\ 1 & -1 & -1 & -1 \\ 1 & 1 & 1 & -1 \\ 1 & 1 & -1 & 1 \end{bmatrix} \begin{bmatrix} \text{Throttle} \\ \text{PID}_{Roll} \\ \text{PID}_{Pitch} \\ \text{PID}_{Yaw} \end{bmatrix} \quad (10)$$

4.3 Proof of Absolute Stability

Fixing Reactive Torque: What happens if the Pitch PID asks for +10 units of force ($\text{PID}_{Pitch} = 10$) while Throttle is 100?

$$\begin{aligned} \Omega_1^2(\text{BR, CW}) &= 100 + (1)(10) = 110 \\ \Omega_2^2(\text{FR, CCW}) &= 100 + (-1)(10) = 90 \\ \Omega_3^2(\text{BL, CCW}) &= 100 + (1)(10) = 110 \\ \Omega_4^2(\text{FL, CW}) &= 100 + (-1)(10) = 90 \end{aligned}$$

Calculate the Yaw torque ($\tau_\psi = \Omega_1^2 + \Omega_4^2 - \Omega_2^2 - \Omega_3^2$):

$$\tau_\psi = (110 + 90) - (90 + 110) = 200 - 200 = \mathbf{0}$$

The math holds: The drone pitches forward with zero induced yaw spin.

Fixing Gyroscopic Precession: If pitching induces a physical gyroscopic precession onto the Roll axis, the independent PID_{Roll} senses this via the gyroscope. Its Derivative term (D) computes a correction, and the Mixer Matrix injects it flawlessly into the motors *without altering the pitch force or inducing yaw*. The axes are 100% decoupled.

5 Conclusion

Idea 1’s diagonal coordinate rotation is mathematically valid in flat geometry (vector superposition), but ignores the 3D physics of rotational rigid bodies. Idea 2 constructs a mathematical bridge (the Mixer Matrix) between abstract physics (Euler angles) and physical actuators (motors). This isolates reactive torque, neutralizes gyroscopic precession, and forms the mathematical backbone of every modern drone flight controller.